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Understanding stakeholders' perspectives on recent initiatives to promote research and development in higher education in Cambodia

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A R T I C L E I N F O

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A B S T R A C T

This study aims to examine the perspectives of relevant stakeholders within Cambodian higher education on recent efforts to promote research and development in higher education in Cambodia. It draws on one-on-one interviews with 13 Cambodian policymakers, educational leaders, lecturers, and university students to collect in-depth qualitative data to understand how these stakeholders perceive the recent initiatives or projects to advance higher education research and development in Cambodia. The findings showed that all stakeholders were generally aware of the progress in research and development in Cambodian higher education; however, some of them could not identify specific initiatives or projects that had recently been introduced to promote research and development. The study highlighted major challenges that have constrained the advancement of research and development in Cambodian higher education, highlighting the challenging realities facing higher education in developing countries. It also offered some suggestions to foster greater research engagement and productivity of academics in Cambodia, which may be applicable to other countries with similar socioeconomic characteristics. The study concludes with a summary of key findings and discusses the study's limitations and implications for future research.

1. Introduction

Higher education plays an essential role in driving socioeconomic development. The role of higher education has been espoused by leading international organisations, such as the World Bank, as a catalyst for developing countries to catch up with their developed counterparts (World Bank, 2002, 2009). In a report titled '*Accelerating Catch Up: Tertiary Education for Growth in Sub-Saharan Africa,*' the World Bank (2009) noted that investment in physical infrastructure and productive facilities is essential; however, if there is a shortage of "technical and managerial skills" and scarcity or low quality of "human capital," economic growth may not be sustained (p. xx). The role of human capital and the ability to produce, use, and apply knowledge have become increasingly crucial as the world is moving deeper into a knowledge-based economy (Heng, 2022).

As Powell and Snellman (2004) noted, "the key component of a knowledge economy is a greater reliance on intellectual capabilities than on physical inputs or natural resources" (p. 199). This suggests that investment in human capital development is "a crucial factor

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that drives productivity growth and raises living standards" (Stiroh, 2000, p. 27). Likewise, as Griliches (1998) emphasized, economic growth depends in part on "the generation of new scientific knowledge in universities and other institutions both at home and abroad, which is itself subject to economic constraint and influences" (p. 2). Inekwe (2015) found that investment in research and development could have a positive impact on GDP per capita in developing countries. Specifically, it was found that a 1% increase in spending on research and development contributes to a 0.06% improvement in economic growth (Inekwe, 2015).

Recognising the role of knowledge in contributing to economic growth, there is a global trend toward a knowledge-based economy. According to Heng (2022), the concept of a knowledge-based economy emerged in the 1970s when the term 'post-industrial society' was introduced by American sociologist Daniel Bell (1919–2011). Along with the trend toward building a knowledge-based economy, there is a growing realisation that knowledge and the ability to produce knowledge have become indispensable, particularly in terms of enhancing innovation capacity in the context of globalisation, rendering knowledge as a key driver of socioeconomic growth (Chankseliani, Qoraboyev, & Gimranova, 2021; Pinheiro & Pillay, 2016). There is also increasing recognition that universities and research-orientated institutions play a pivotal role in knowledge generation (Lovakov, Chankseliani, & Panova, 2022), which in turn contributes to promoting competitiveness in the global knowledge-based economy where the capacity to produce, utilise, and disseminate knowledge is key for economic development (Chankseliani, Fedyukin, & Froumin, 2022; Heng, 2022; Pinheiro & Pillay, 2016). Thus, the role of universities and higher education institutions (HEIs) has gained prominence in recent decades, prompting many countries, including those in the Global South, to build their universities' research capacities (Chankseliani et al., 2022; Postiglione & Arimoto, 2015), and develop "regional spaces that transcend national boundaries and foster cross-border collaboration and integration" (Chankseliani & Sopromadze, 2023, p. 2).

In contemporary Cambodia, considerable efforts have been made to promote higher education and university research as the country aspires to become a knowledge-based society (Heng, 2023a; MoEYS, 2014). Since the late 1990s when the privatisation of higher education was introduced, Cambodian higher education has experienced a remarkable transformation (Un & Sok, 2018). This transformation saw public higher education institutions (HEIs) beginning to offer fee-paying degree programs and private HEIs starting to mushroom. In the 1990s, there were fewer than 10 HEIs and around 10,000 students in higher education in Cambodia (Un & Sok, 2018). By 2005, the number of HEIs rose to 51, up from only eight in 1997 (Chet, 2009). In the latest Education Congress report (MoEYS, 2023), there are 132 HEIs, 84 of which are privately owned. These 132 HEIs serve a population of 209,059 students (45.7% females) and employ 16,471 education staff (about 20% females), of which 23.59% hold bachelor's degrees, 67.57% hold master's degrees, and only 8.83% hold doctorates (MoEYS, 2023). All HEIs in Cambodia are supervised by 16 different ministries and state institutions, which creates an issue of fragmentation in higher education governance in the country (Mak et al., 2019).

With the rapid expansion in the number of HEIs and student enrolment that increased from around 10,000 in the late 1990s to around 200,000 in the early 2020s (Heng et al., 2022), the quality of higher education has become a major issue (see MoEYS, 2021; Poeu, 2017; Un & Sok, 2018). For example, research has consistently shown that Cambodian higher education is beset with low quality. Many university students graduate with skills mismatches - their skills are not in alignment with the market needs (Heng & Sol, 2023; Madhur, 2014; Peou, 2017). There are also challenges related to the cultural norms and socioeconomic circumstances of Cambodia which made it hard for Cambodian higher education to navigate the demands of globalization (Howes & Ford, 2011; Sen, 2019). In addition, the legacy of the previous form of higher education (i.e., the Soviet legacy) may have influenced the development of the current higher education in Cambodia (Oleksiyenko, 2021). This is not to mention the legacy of the prolonged civil war and the previous regimes (Sen, 2019), as well as the aftermath of the genocidal Khmer Rouge regime (1975–1979), which resulted in "a near-total loss of an entire generation of academics" (Kwok et al., 2010, p. 35). These events forced Cambodia to rebuild its research culture from scratch.

Moreover, there is a noticeable lack of research activities. For example, a relatively large-scale survey by Eam (2015) involving 444 lecturers from 10 Cambodian universities showed that about 65% of Cambodian lecturers were not engaged in research at all. Other studies by Kwok et al. (2010), Oleksiyenko and Ros (2019), and Heng, Hamid, and Khan (2022, 2023b) have also shown limited research involvement amongst Cambodian faculty members. For example, while it was found that the level of research engagement of Cambodian academics in public universities was limited, that of academics in Cambodian private universities was "almost completely absent" (Heng et al., 2023b, p. 330).

The limited research involvement and capacity of Cambodian HEIs and lecturers have prompted MoEYS to introduce policies and projects that aim to promote research activities in Cambodian higher education (Heng, 2023b). In recent years, two key projects have been implemented. The first project is the Higher Education Quality and Capacity Improvement Project (HEQCIP) which lasted from 2010 to 2017. The project, valued at US$23 million, received a loan from the World Bank, Cambodia's key development partner. The other project is the Higher Education Improvement Project (HEIP). It is also funded by a World Bank loan, spanning six years from 2018 to 2024. HEIP is valued at US$92.5 million and aims to promote teaching and research in the fields of STEM (science, technology, engineering, and mathematics) and agriculture (see Heng & Sol, 2021 for further discussion). Thus far, little research has provided an assessment of these projects. With the exception of MoEYS (2015), Rappleye and Un (2018), and Heng (2021, 2023b), there appears to be a lack of research investigating the impact of these projects and how they are perceived by concerned stakeholders in Cambodian higher education.

This study aims to fill this research gap and contribute to a better understanding of how recent initiatives to promote research and development in higher education in Cambodia are perceived by relevant stakeholders, such as policymakers, educational leaders, faculty members, and university students. Such an understanding will be useful for evidence-based policy formulation and implementation. The study is guided by one research question: How do higher education stakeholders in Cambodia perceive the recent initiatives to promote research and development in Cambodian higher education?

1.1. Global initiatives to promote research and development

The last few decades have seen global efforts to promote research and development, particularly in industrialised and emerging economies. According to Shore and Wright (2016), universities across the world have experienced reforms and reconfigurations to ensure that they can effectively respond to the ever-changing demands of contemporary societies. The reforms or efforts to enhance the roles of universities are motivated by the increasing realization of the roles of knowledge in supporting socioeconomic growth (UNESCO, 2022). As Altbach (2008) argued, universities are "public-good institutions" that play an essential role in knowledge-based societies (p. 13). They are fountains of knowledge and help to produce "the world's most important resources: young minds and an educated labour force" that can contribute to socioeconomic development (Breznitz, 2014, p. 1). Moreover, there has been growing global competition and collaboration in scientific research, which drives research and development in higher education across different regions (Chankseliani, 2023; Jung & Horta, 2015; Lovakov et al., 2022; Powell et al., 2017). This trend has pushed universities and research institutions to strive for greater levels of innovation and quality in their research outputs and diversify funding sources by seeking various grants, collaboration, and partnerships (Baker, Dusdal, Powell, & Fernandez, 2018). The changing landscape of national policies has also played a crucial role in fostering higher education research and development in many countries, as they envision becoming a knowledge-based society (see Chankseliani, 2023; Powell et al., 2017; UNESCO, 2022).

Given the crucial role of universities, there are efforts introduced in different higher education contexts to promote research and development, a crucial factor for enhancing national innovation capacity (Marginson, 2010). For example, to follow their counterparts in North America and Europe, countries in Asia have invested in building their universities to transform them into world-class universities (Shin & Kehm, 2013). Specifically, China has introduced Project 211, Project 985, and the Double First-Class University Plan to strengthen the research capacity of its universities (Cao & Yang, 2023). Other East Asian countries, such as Japan and South Korea, have also introduced similar initiatives to enhance the research performance of their universities (Shin & Kehm, 2013). South Korea, for instance, has implemented various projects, such as Social Science Korea and World-Class University, to promote research and development and build world-class universities (Shin & Jang, 2013). Many Southeast Asian countries, including Singapore, Thailand, Vietnam, and Malaysia, have made parallel efforts to advance research and development within their HEIs and research institutions (see Dhirathiti, 2018; Hooi & Wang, 2020; Jung & Horta, 2015; Nguyen, 2020; Sheriff & Abdullah, 2017). Singapore, for instance, has persistently positioned itself as a regional and global leader in innovation and research through its multifaceted approaches, including substantial government funding, high-quality research infrastructure, strategic partnerships with top universities worldwide, and a focus on aligning research with its national goals (Hang, Low, & Thampuran, 2016; Hooi & Wang, 2020; Sidhu, Ho, & Yeoh, 2011). Notably, the Research, Innovation, and Enterprise (RIE) Plans 2011–2015, 2016–2020, and 2021–2025 have collectively allocated S $60 (about US$44) billion to drive research and development in the country (National Research Foundation Singapore, 2020).

Similar efforts are also evident in Latin America. In particular, the National Scientific and Technical Research Council (CONICET) of Argentina has actively funded research projects, including those from universities; enhanced the research infrastructure and resources of numerous research institutes across the country; and collaborated with national and international organisations, universities, and industry to promote research, technology transfer, and innovation (Marquina & Luchilo, 2021). Likewise, many other Latin American countries, such as Brazil, Columbia, and Mexico, have undertaken endeavours to advance their higher education research and development. Brazil, for example, has promoted research and development in its higher education through initiatives such as the Science Without Borders (SWB) program, research grants for students and researchers, and university-operated science parks and technology incubators (Dalmarco, Hulsink, & Blois, 2018; Nery, 2018).

Countries in Africa have demonstrated parallel efforts to boost higher education research and development through a number of key research-focused initiatives such as the Study Programme on Higher Education Management in Africa, the African Research Universities Alliance (ARUA), African Research Initiative for Scientific Excellence (ARISE), Africa Research Excellence Fund (AREF), and the Alliance for Accelerating Excellence in Science in Africa (AESA) (Ogega, 2023; Ramutsindela & Mickler, 2020; Sawyerr, 2004; Taylor-Robinson et al., 2022). Moreover, several African governments and international development organisations have supported the establishment of Centres of Excellence in different disciplines to enhance the quality of postgraduate education and foster higher education research and development in Africa (World Bank, 2020).

Despite global initiatives and efforts aimed at promoting research and development within various higher education contexts, persistent challenges rooted in sociocultural, economic, and political factors continue to influence higher education practices and progress (Howes & Ford, 2011; Jabnoun, 2009; Lee & Kuzhabekova, 2019; Sen, 2019, 2022). Specifically, sociocultural factors, such as societal perceptions of education, local values, gender norms, and other deeply ingrained beliefs and traditions, can either promote an environment conducive to educational innovation or, on the flip side, serve as challenges to such development (Jung & Horta, 2015; Lee & Kuzhabekova, 2019). For instance, in societies where educational innovation and scientific achievements are strongly valued, there is often substantial investment in research and development by channelling significant public and private funding into research-intensive universities or institutions (Soete, Verspagen, & Ziesemer, 2022). Economic factors, which can significantly influence financial capacities, also play a crucial role in the research and development landscape within higher education (Jabnoun, 2009; Shin, 2012). It is widely acknowledged that financial resources are fundamental to support higher education research and development in various aspects, including, amongst others, funding for research projects, development and upgradation of research infrastructure, recruitment and retention of highly qualified faculty and researchers, and participation in international collaborations (Minguillo & Thelwall, 2015; Nguyen, 2016). Pressures from economic conditions can potentially skew priorities towards applied research at the expense of basic research, limit institutional ability to invest in state-of-the-art research facilities and equipment, place HEIs at a competitive disadvantage in terms of recruiting and retaining top talent, and impede their ability to engage in international partnerships and collaborations. Political factors also exert a substantial influence on higher education research and development. In particular, political decisions determine how much expenditure or investment is allocated to higher education and establish regulations and policies that directly shape how HEIs operate (McLendon, Hearn, & Mokher, 2009; Tandberg & Ness, 2011). One of the widely discussed issues is academic freedom (Altbach, 2007; Rostan, 2010). Political views or ideologies of those in power, especially those with authoritarian tendencies, might suppress research that opposes their policies or political agendas (Berggren & Bjørnskov, 2022; Williams, 2006). This circumstance can lead to self-censorship amongst academics and researchers due to fear of repercussions, which is severely detrimental to the development of innovative ideas and new discoveries necessary to boost research and development (Kaufmann, 2021; Rostan, 2010; Un & Sok, 2018).

1.2. Initiatives to promote research and development in Cambodian higher education

Recognising the role of research in generating new knowledge to tackle real-life problems and drive sustainable socioeconomic development through innovation, breakthroughs, and evidence-based policies, the Royal Government of Cambodia, through MoEYS, has introduced a series of research-related policies, including the Policy on Research Development in the Education Sector, the Master Plan for Research Development in the Education Sector 2011–2015, the Policy on Higher Education Vision 2030, and the Higher Education Sub-Sector Strategy 2021–2030 (Heng & Sol, 2021). With support from its long-standing development partners, especially the World Bank, MoEYS has also invested considerably in developing higher education research in Cambodia, particularly since 2010. As previously mentioned, two major higher education projects, HEQCIP and HEIP, have been introduced. One of the key objectives of the two projects is to provide enabling conditions for the development of research activities and research quality in Cambodian higher education, with HEIP mainly focusing on research in STEM and agricultural fields at five targeted public HEIs. In quantitative terms, HEQCIP funded 45 research projects under its 'Development and Innovation Grants' scheme of over US$3.5 million (MoEYS, 2015), while HEIP has funded 53 research projects (worth US$15.8 million) that will mostly be completed by 2023 (World Bank, 2021).

In 2020, MoEYS unprecedentedly announced two more major initiatives: The Research Creativity and Innovation Fund (RCI Fund) and a proclamation on the recruitment and appointment of professors in the education sector (Heng, 2020). The RCI Fund aims to cultivate a research-orientated culture, stimulate creativity, and promote innovation in response to the evolving labour market and globalisation (MoEYS, 2020b). The RCI Fund provides financial support for research in three prioritised areas: (1) digitalisation for Industry 4.0, (2) applied agriculture, and (3) 21st-century pedagogy. By the same token, MoEYS has implemented the professorial ranking policy, which has provided a booster for higher education research and publication in Cambodia. As outlined in the proclamation on the appointment of professors, to be qualified for each professorial title (Assistant Professor, Associate Professor, and Professor), a certain number of academic publications and conference presentations, in addition to other criteria, are required (Heng, 2020). Specifically, the publication of seven journal articles and two books is required for the title of Professor, while the titles of Associate Professor and Assistant Professor require five journal articles and one book, and three journal articles or one book, respectively (MoEYS, 2020a). Moreover, the recent establishment of several national and institutional research forums and conferences (e.g., National Research Forum, Cambodian ELT Conference) and Cambodia-based academic journals (e.g., *Cambodian Journal of Educational Research, Insight: The Cambodia Journal of Basic and Applied Research, Cambodian Journal of Education and STEM*) has also marked a significant development in the research and higher education landscape in Cambodia in recent years (see Heng & Sol, 2021; Ros & Heng, 2022).

Nonetheless, with a few notable exceptions (e.g., Heng, 2021; MoEYS, 2015; Rappleye & Un, 2018), few studies have examined the impact of these recent initiatives in promoting research and development in Cambodian higher education and explored how higher education stakeholders in Cambodia perceive these recent research-promoting initiatives. Therefore, this study aims to address this knowledge gap and contribute to a better understanding of the new developments in research and development in Cambodian higher education. Such an understanding will be useful for policymakers and concerned stakeholders to enhance policy formulation and implementation in order to promote research and development in Cambodian higher education and the nation at large.

2. Research methods

2.1. Research design

This study was designed as a qualitative enquiry (Yin, 2015), informed by a social constructivist view of knowledge creation which postulates that knowledge and realities can be socially constructed by the dynamic interactions between researchers and participants (Bryman, 2012). The aim of the study was to examine in depth stakeholders' perspectives on the recent initiatives to promote research and development in Cambodian higher education; therefore, a qualitative approach to enquiry was deemed suitable to achieve this aim.

2.2. Research setting and participants

This study took place in Phnom Penh, the capital city of Cambodia. It involved a total of 13 participants who were key stakeholders in higher education in Cambodia. To ensure that a variety of perspectives are represented, we selected a few participants from each stakeholder group such as policymakers, educational leaders, university lecturers, and university students (bachelor's, master's, and doctorate degrees). In selecting the participants for the study, we used a purposive sampling strategy (Patton, 2015) and considered their position, gender, institutional type, educational background, discipline, and work experience. This resulted in the selection of 13 participants (coded as K1-K13): two policymakers, one educational leader, four university lecturers, and six university students. These participants were selected from nine different institutions: one ministry (i.e., MoEYS), three public universities (Public university A-C), and five private universities (Private university A-E). Amongst the 13 participants, four were females, five held a doctoral degree, and two were from science disciplines. Their work experience ranged from six months to 24 years, and their ages were between early 20 and early 50. The profiles of the interview participants are provided in Table 1.

2.3. Data collection and analysis

To collect the data for analysis, semi-structured interviews were used. The interviews were conducted in Khmer, the native language of both the researchers and the participants. All interviews, except for one (K1) that was conducted face-to-face, were carried out via Zoom. The interviews were conducted in early 2022 and lasted between 30 and 60 minutes and were recorded with the participants' informed consent. The recorded interviews were transcribed and translated into English and were checked by both researchers for accuracy in meanings and language use. Any ambiguous points were reviewed, discussed, and finalised before the data analysis began.

To analyse the data, we used qualitative data analysis software NVivo 12, as this software can effectively facilitate data storage, coding, and analysis. Both researchers were involved in the data collection, coding, and analysis processes. The data analysis followed Braun and Clarke's (2006) thematic analysis which consisted of six steps such as (1) familiarising oneself with the data; (2) generating initial codes; (3) searching for themes; (4) reviewing themes; (5) defining and naming themes; and (6) producing the research report.

In this study, the participants are identified by codes: K1–K13 (K = Key informant). This is to protect their identities and comply with ethical standards in scientific research. To enhance the credibility and trustworthiness of this research, we adopted several strategies to enhance research rigour suggested by Houghton et al. (2013). In particular, we used the member-checking technique, triangulated the interview data with our experience and knowledge of the research topic, and practised reflexivity to mitigate researcher bias in the research process. Our analysis revealed three major themes, including (1) Recent progress in research and development in Cambodian higher education, (2) Challenges to research and development in Cambodian higher education, and (3) Suggestions to promote research and development in Cambodia.

3. Findings

3.1. Recent progress in research and development in Cambodian higher education

The data analysis revealed that almost all of the participants agreed that there was some noticeable progress in research and development in Cambodian higher education. While some participants could describe the recent initiatives or projects to promote research, such as HEQCIP and HEIP, others were unaware or vaguely aware of them. Key recent developments to promote research activities that were most frequently mentioned by the participants are summarised in Table 2.

As seen in Table 2, major developments in research and development in Cambodian higher education taking place in recent years included (1) the introduction of research-promotion policies and projects, (2) an improvement in the research ecosystem, (3) the allocation of research funds/grants, (4) the introduction of research-focused curricula, and (5) an increase in research and publication activities. The following quotes highlight these recent developments in research in Cambodian higher education:

Recently, MoEYS has considered introducing the policy on academic ranking (assistant professor, associate professor, and professor). To be appointed a professorial title, lecturers must have a good record of research activities and publications in addition to advanced university degrees. (K7)

The research course at my university takes one year, providing step-by-step guidance on how to do research. … students need to read a lot of research articles and write summaries. (K9)

Another participant shared his observation about the increase in international publications amongst Cambodian universities:

There has been a noticeable increase in the number of publications in international journals from both public and private

Table 1
Profiles of the interview participants.

Participant code	Type of participants	Gender	Institution	Degree	Discipline	Work experience
K1	Policymaker	Male	MoEYS	PhD	Education	18 years
K2	Policymaker	Female	MoEYS	PhD	*	14 years
K3	Educational leader	Male	Private university A	PhD	Education	24 years
K4	Full-time lecturer	Male	Public university A	Master	Language Education	15 years
K5	Full-time lecturer	Female	Public university B	PhD	Chemical Science and Engineering	2.5 years
K6	Part-time lecturer	Male	Public university C	PhD	Veterinary Medicine	3 years
K7	Part-time lecturer	Male	Public university A	Master	Language Education	4 years
K8	PhD student	Male	Private university B	Master	Language Education	15 years
K9	PhD student	Male	Private university C	Master	Language Education	7 years
K10	Master's student	Male	Private university D	Master	Education & Business Management	17 years
K11	Master's student	Male	Public university A	Bachelor	Education	5 years
K12	Bachelor's student	Female	Private university A	Bachelor's student	English	6 months
K13	Bachelor's student	Female	Private university E	Bachelor's student	International Relations	6 months

* The participant's discipline has been anonymised to protect her identity.

Table 2
Key recent developments that promote research activities in Cambodian higher education.

Key recent developments	Number of participants	Number of mentions	Exemplary quotes
Introduction of research-promotion policies and projects	12	44	• Key research development in Cambodian higher education is the introduction of HEQCIP and HEIP. (K1) • There are various research-related policies not only for higher education but also for general education. …some major universities started to establish Centres for Research. (K2)
Improvement in the research ecosystem	10	25	• With regard to human resource development, more researchers get PhDs from overseas and have a good record of publications. (K1) • HEIP has improved the research ecosystem. …more funding to improve research infrastructure and lab equipment. …here we promote a laboratory-based education (LBE) system and university-industry linkages. (K5)
Allocation of research funds/grants	10	10	• The World Bank and ADB have provided more funding to implement research projects. (K3) • At my university, there are research funds for competition ranging from \$5000 to \$10,000 per project. (K4)
Introduction of research-focused curricula	8	24	• At my university, undergraduate students take a one-year research course. (K7) • Some universities require master's and PhD students to conduct research as part of the graduate requirements. (K10)
Increase in research and publication activities	8	20	• … increased number of publications in both local and international journals amongst Cambodian researchers and academics. (K5) • More lecturers are involved in research activities, especially from the Department of International Studies. (K7)

Table 3
Key challenges preventing the development of research and development in Cambodian higher education.

Key challenges	Number of participants	Number of mentions	Exemplary quotes
Limited research knowledge and skills	10	27	• One more challenge is the limited research capacity amongst researchers and academics. (K5) • Some advisors do not have publication experience, so it is hard for them to supervise research students. (K8)
Lack of research funding, grants, and incentives	10	26	• [There is] limited financial support, so academics choose to teach many classes to earn more income. (K3) • There is a general lack of funding for research activities. If there is, only small funding is available. (K7)
Lack of leadership commitment and capacity	10	21	• It is also related to leadership. They might not know about research. Leaders don't just know about management. They need to have technical skills. What are the roles of their universities? If they don't have that knowledge, they cannot do it. (K1) • Research development is only in the mouth of leaders, no committed actions. (K7)
Insufficient time for research activities	11	17	• Cambodian universities focus too much on teaching (teaching-orientated). Academics want to teach more classes, so they don't care about doing research. (K2) • Many academics focus on consultation and other income-generating activities, so research and development won't grow. (K4)
Lack of clear national and institutional research policies	5	12	• We don't have a promotion system based on research performance. We have a professorial ranking, but it is pending.… There is no system in place to lead them to do research. (K1) • Research policies tend to be ambitious, so cannot be implemented fully. (K11)
Poor mindset towards research	5	8	• Research is not strongly valued. It is the current mindset or attitudes towards research in our country. (K2) • People don't see the value of research and how it can influence policies. (K13)
Language barriers	4	5	• Language is a barrier to research as we need a good level of English. (K6) • Most research is conducted and published in English. This makes it harder to conduct research. (K13)
Lack of academic freedom	3	4	• There is political motivation in research. Some government officials, when interviewed or surveyed, always speak highly of the work of the government, so we cannot find real problems. Some sensitive topics are also prohibited from research. (K3) • … no complete freedom for research, for example, topic selection or publication decision. We need to give the report to them and cannot publish it elsewhere. (K7)
Limited access to research data and resources	4	4	• … high bureaucracy in terms of paperwork and decisions on research permission, especially when the research involves public institutions. (K3) • Universities do not have subscriptions with journal and book publishers. It is hard to have access to good journal articles or books. (K7)

universities. (K1)

Several other participants (e.g., K3, K4, and K6) mentioned the availability of funds for research projects. For example, K6 said that:

There are funds for lecturers and students to implement research projects, for example, funds from HEIP and CE SAIN (Centre of Excellence on Sustainable Agricultural Intensification and Nutrition).

Although many participants were aware of the progress in research and development in Cambodian higher education, most of them could not identify the specific initiatives that were introduced by MoEYS, their universities, and other HEIs. This indicates that in general the participants could observe the change or improvement in the research landscape in Cambodia, but they did not clearly know the specific projects, initiatives, or activities that had been introduced. The following quotes illustrate this point:

I have heard about these initiatives (HEQCIP and HEIP), but I am not sure of what they are all about. (K3)

I don't know about these projects (i.e., HEQCIP and HEIP). (K8)

I have never heard about these projects (i.e., HEQCIP and HEIP). (K13)

Nonetheless, the findings revealed a consensus amongst the participants that research has improved in recent years despite the many challenges that continue to impede the development of research and development in Cambodian higher education and Cambodia as a whole. As several participants pointed out during the interviews, they could see more research and publication activities within the past several years. Some went on to highlight the establishment of new academic journals, such as *Insight: Cambodia Journal of Basic and Applied Research* published by the Royal University of Phnom Penh and *Cambodian Journal of Educational Research* published by the Cambodian Education Forum, just to name a few.

Overall, this study showed that there were some positive developments in research and development in Cambodian higher education. Key developments were seen through the various initiatives aimed at promoting research and the overall increase in research and publication activities (see Table 2). This finding indicates that the efforts by the Education Ministry and relevant stakeholders, particularly universities, international development partners, and think tanks, have been fruitful, suggesting a somewhat positive outlook for the future of research and development in Cambodia.

3.2. Challenges to research and development in Cambodian higher education

In spite of the progress in research and development over the past decade, the participants highlighted many persistent challenges that had prevented and will continue to prevent the progress of research and development in Cambodian higher education. Key challenges that were frequently mentioned centred around academics' inadequacy in research knowledge and skills; limited availability of research funds, grants, and incentives; lack of research-orientated leadership; insufficient time for research activities; lack of clear national and institutional research policies; unfavourable mindset or attitudes toward research; language barriers (i.e., limited English writing skills); lack of academic freedom; and limited access to research data and resources. These challenges are shown in Table 3 with exemplary quotes from the participants.

As Table 3 shows, there are various challenges to the advancement of research and development in Cambodian higher education. Amongst them, a few issues received the most mentions by the participants. They included: (1) limited research knowledge and skills; (2) lack of research funding, grants, and incentives; (3) lack of leadership commitment and capacity; and (4) inadequate time for research activities and supervision. As one participant noted:

In our higher education context, many lecturers lack research knowledge and skills, so most of them can only publish in local journals.… There is a research coordinator in my department, but the person seems not really capable of leading and pushing research activities amongst lecturers. (K7)

Another participant who was a doctoral student had a similar observation. He said:

Those who know about research and who are committed to research are few, so the development of research and development remains limited. (K8)

In addition, the participants also mentioned other challenges that they considered to be inhibiting factors for research and development in Cambodia. These included a lack of research-orientated leadership, limited sources of motivation for research, a laissez-faire approach toward research, and heavy teaching loads. The following quotes illustrate these challenges, with key comments highlighted in italics:

There is not much effort towards research at the institutional level. It is about leadership. *They are not researchers*. Leadership is crucial. (K1)

We need financial support to conduct research. Some research, especially STEM research or research with a large sample for policy formulation, requires a lot of funding. However, *the allocation of public funding for research remains very limited. Financial and non-financial benefits* for academics or researchers are also limited. (K2)

I think research and development in Cambodia will take more time. I have been in the higher education sector for more than 20 years. I haven't seen much progress so far. *There has not been sufficient push both from the institutional and governmental levels*. (K3)

Lecturers teach too many courses, so *they don't have time for research activities*. (K12)

Taken as a whole, the challenges to research and development in Cambodian higher education described by the participants were mainly linked to several key barriers, including the inadequacy of knowledge and capacity to conduct research, lack of financial support for research, and lack of effective leadership that drives research activities. These barriers, coupled with other challenges such as the lack of academic freedom and clear policies to promote research, have created an environment that is far from conducive for research and development to flourish.

3.3. Suggestions to promote research and development in Cambodia

The data analysis pointed to several key suggestions that the participants believed would promote research and development in Cambodian higher education in particular and Cambodia in general. The two policymakers from the Education Ministry who were interviewed for this study stressed the importance of introducing specific research requirements for university lecturers. One of them said:

Cambodian universities should make it mandatory for their lecturers to conduct and publish research as part of their performance appraisals. (K2)

These policymakers also believed that the provision of research incentives, funds, or grants was important. As articulated by one of them:

We need both public funding and institutional budgets to sustain research activities. We also need to make sure that they (academics/researchers) can get financial benefits from doing research. (K1)

In addition, they suggested providing research training for lecturers. For example, K1, who had worked closely with university lecturers on a number of research projects, said, 'it is vital to equip lecturers with more research skills.' He added that, through HEIP, there was an ongoing mentoring program in which researchers were mentored by their more experienced colleagues so that they could complete their research and get it published successfully.

The educational leader (K3) who was a faculty dean and had more than 20 years of experience in higher education was rather pessimistic about the current status quo of research in Cambodian higher education. His suggestions focused on creating a conducive environment for research. He stressed the need to allow for better academic freedom in addition to supporting researchers by reducing bureaucracy and increasing the availability of research funds and grants. The following quotes illuminate his suggestions:

Academic freedom needs to be promoted. Let them research any topics to find problems and suggest solutions.

Relevant authorities need to improve paperwork procedures, allowing students, academics, and researchers to fulfil their ethical requirements and access the necessary data and research samples.

MoEYS should allocate enough research funding to higher education institutions. Funding is a force to drive research activities and involvement. (K3)

The lecturers who participated in this study raised major issues concerning the lack of funds and grants for research as well as the absence of research incentives in the form of career progression or additional wages or salary resulting from their performance in research and publication. Their suggestions, therefore, focused on the provision of research grants, funds, or incentives. Further, they also suggested providing research training, introducing clear research policies, improving the quality of the institutional research office, providing more academic freedom, and allocating more time for research activities. Their suggestions can be summarised into five key points, as shown in Table 4.

As can be seen in Table 4, the lecturers who participated in this study believed that more research capacity-building opportunities should be provided to develop and strengthen the research knowledge and skills of Cambodian lecturers, many of whom did not receive rigorous research training that prepared them to be researchers. As one participant put it:

Universities cannot push research forward while lecturers lack research knowledge and skills. They need to provide lecturers with capacity-building opportunities, such as research skills training and mentoring and coaching with more experienced peers. (K6)

The lecturers also stressed the need to provide incentives, grants, or funds for research to encourage research engagement activities amongst Cambodian academics. This suggestion is practical and understandable, given the low academic salaries for both public and private university lecturers. As K4 said, funding support is necessary to drive research interest and activities. He added that there should be incentives for academics who publish research and help build their university's reputation.

Other suggestions from the lecturers focused on establishing and implementing clear research policies that outline how research is required and incentivised, providing sufficient resources and time for research, and creating a conducive environment for research (see Table 4).

As for the university students, they shared several key suggestions, some of which overlapped with the above suggestions made by other participants. However, as students, they tended to focus their suggestions on their universities and lecturers. K8 and K9, who

Table 4
Key suggestions to promote research made by the lecturers.

Key suggestions	Exemplary quotes
Provide research capacity building	• Universities should offer PhD training to lecturers who don't have a doctorate to upgrade their research knowledge and skills. (K4) • Lecturers need more training in research skills. (K6)
Provide research incentives, grants, or funds	• Both financial and non-financial incentives should be provided to those involved in research activities. (K6) • More research funding should be made available with a transparent process. (K7)
Establish clear research policies	• Universities need to have a clear policy framework to promote institutional research. (K4) • A clear M&E (Monitoring and Evaluation) policy for institutional research needs to be put in place. (K5)
Provide sufficient research resources and time	• More books in Khmer and research facilities are needed. (K6) • Universities should allocate enough time for lecturers to engage in research activities. (K7)
Create a conducive environment for research	• Universities should employ highly experienced technicians and researchers to support and implement research activities. (K6) • The quality of offices or persons in charge of research needs to be improved. (K7)

were PhD students, for example, pointed out that the research capacity of lecturers should be a priority for Cambodian HEIs.

Universities should recruit lecturers with strong research backgrounds and let them lead student research. (K8)

A priority for Cambodian universities is to build up the research capacity of their lecturers and make sure that they supervise students and lead research within their field of expertise. (K9)

K8 added that universities should require master's and doctoral students to publish at least one research article as part of their graduation requirements. He agreed that although some universities now require their PhD students to publish at least one article in local or international journals, many do not have this requirement. Others tend not to care much about the quality of publications, accepting any publications including those published in predatory journals.

K10 and K11, who were master's degree students, shared the following suggestions:

A centre for research should be established in each university to provide research guidance to students and run various research training, workshops, or seminars. The research centre should also try to form research collaborations with both local and international institutions.… A clear institutional research policy for lecturers and students should be put in place. (K10)

Universities should enhance their research courses, providing students with practical research knowledge and skills.… and encourage them (students) to engage in research activities, such as attending or presenting papers at research conferences. Research funding should be made available for researchers, lecturers, and students.... Universities should also allocate enough time for lecturers to supervise student research as well as be involved in research activities.... There should also be an incentive system to recognise research outputs and efforts made by lecturers. (K11)

As for the two bachelor's degree students, they tended to direct their suggestions to their lecturers and universities. One of them suggested that universities should introduce the research course earlier in their curricula, preferably in the first or second year. The other stressed the need to recruit lecturers with strong research experience so that they can support students in grasping practical research knowledge and skills. The following quotes underscore their key suggestion points:

The research course should be introduced earlier, like in Year 1 or 2, so it can help students to complete assignments in later years. (K13)

Universities should be selective when recruiting lecturers. They need to focus on the quality and research experience of the lecturers.… They (lecturers) need to support students in doing and publishing research. (K12)

Overall, the participants in this study offered various suggestions that can be classified into four key points: (1) establishing clear research policies, (2) providing research incentives or funds, (3) offering training and mentoring support, and (4) creating a conducive environment for research. Acting on these suggestions will pave the way for more interest in research, which will in turn have the potential to enhance research activities and productivity of Cambodian academics and universities.

4. Discussion

4.1. Progress in the face of considerable challenges

The findings of this study show that there has been some noteworthy progress in the realm of research and development in Cambodian higher education in recent years, corroborating observations noted by Heng (2020, 2023b) and Heng and Sol (2021). The recent progress in research and development reflects the need for Cambodia to catch up with the regional and global trends in higher education and research and development around the world (Baker, Dusdal, Powell, & Fernandez, 2018; Jung & Horta, 2015; Powell et al., 2017). As noted in its Education Strategic Plan 2019–2023, the Education Ministry has a mission to develop the education system in Cambodia in response to social, economic, and cultural development needs as well as "the reality of regionalisation and globalisation" (MoEYS, 2019, p. 18). It also aims to improve the quality and relevance of its education system, particularly higher education, to ensure that Cambodian graduates are equipped with the necessary knowledge and skills that enable them to meet regional and international standards and become competitive in the global job markets (MoEYS, 2019).

The recent progress in research and development in Cambodia, observed in this study, reflects the need for the country to stay relevant and competitive on the regional and global stage, driven by various internal and external forces. Internally, post-war political stability and the need for visible and sustainable socioeconomic growth have motivated the Cambodian government to introduce a number of structural reforms and initiatives, including those in the education sector (see McNamara & Hayden, 2022; Sen, 2019), to drive social, economic, and educational development. This is evidenced by the implementation of HEQCIP, HEIP, and other higher education initiatives (Heng et al., 2023a), with an overall aim to enhance sectoral governance, improve the quality of teaching and research, and promote research in Cambodian higher education (World Bank, 2021). However, such projects and initiatives are predominantly funded by international development partners, as seen in other countries where local funding for research is limited (Chankseliani, 2023). There are also external forces, including globalization and the internationally growing trend toward a knowledge-based economy, which have profoundly influenced the research and development landscape in Cambodia (Heng, 2023a) and other developing countries (Powell et al., 2017; UNESCO, 2022).

Despite the progress, it is important to note that such recent progress and developments are taking place in the face of considerable structural challenges, rendering the future outlook for research and development in Cambodian higher education bleak. These findings are consistent with previous research investigating academic life (Chhaing, 2022), research conceptions (Heng et al., 2022; Nhem, 2023), and academic research development in Cambodia (Heng, 2023; Heng et al., 2023b). For example, Chhaing (2022) has revealed a common desperate need amongst many Cambodian academics to get out of their teaching profession and seek employment elsewhere due to unfavourable academic life, undesirable working conditions, and network-based, unstable employment practices. Heng et al. (2023b) have shown that Cambodian academics "operate in a precarious environment characterised by low salaries, inadequate institutional support, and lack of incentives and recognition for research" (p. 332). Overall, the challenges facing higher education research and development in Cambodia are rooted in the broader aspects of its sociocultural, economic, and political contexts (Howes & Ford, 2011; Kwok et al., 2010; Ros, Eam, Heng, & Ravy, 2020; Sen, 2019, 2022).

The challenges facing Cambodian academics are, however, not unique to the Cambodian setting. Previous studies about research and higher education development in other developing contexts have revealed somewhat similar issues and challenges. For example, research from China (Bai, Millwater, & Hudson, 2013), Ghana (Abugre, 2017), Ukraine (Hladchenko, de Boer, & Westerheijden, 2016), Tanzania (Fussy, 2019), Thailand (Rungfamai, 2018), Vietnam (Nguyen, 2013), and post-Soviet countries (Chankseliani et al., 2022) have shown that academics in these contexts faced a range of barriers stemmed from insufficient human, physical, financial, and technical resources, which characterise the unique and unfavourable realities confronting academics in the Global South (Altbach, 2003; Canagarajah, 2002; Fussy, 2019; Horta & Mok, 2020; Sawyerr, 2004). Thus, the findings of the present study not only attest to the geopolitics of knowledge production (Canagarajah, 2002; Chankseliani, 2023) but also provide a nuanced understanding of the challenging realities of academic life in a developing country like Cambodia.

4.2. The need to create a conducive environment for research

The findings of this study highlight the need to create a conducive environment for research in Cambodian higher education and other similar higher education contexts. As discussed above, the participants made various suggestions to promote research and development in Cambodia (see Table 4). Key suggestions seem to centre around introducing research requirements, providing research incentives and training, improving academic freedom, establishing a clear research policy, allocating time for research, and improving the function of institutional research offices. These suggestions are well-founded, given the challenging realities facing Cambodian academics, hindering their active engagement in research. For example, Heng et al. (2023b) have highlighted the various structural constraints to academic research in Cambodia. They noted that Cambodian academics "operate in a precarious environment characterised by low salaries, inadequate institutional support, and lack of incentives and recognition for research" (p. 11). The findings of the present study show that the above statement is true, thereby suggesting a strong need to introduce change to create a conducive environment for research.

The policymakers from the Education Ministry who participated in this study suggested the need to introduce research requirements and provide research incentives, funds, and grants to encourage Cambodian academics to engage in research. These suggestions are in line with what was discussed by previous research which found that the lack of requirements and incentives for research were amongst the major barriers to research development in Cambodia (Heng et al., 2022, 2023b) and other developing countries (Fussy, 2019; Horta & Mok, 2020). In addition, as the issue of limited academic freedom is another key challenge to research in Cambodian higher education in particular and Cambodia in general, it is crucial to ensure that this fundamental structural barrier, alongside other challenges and inhibiting factors as discussed by the Cambodian Institute for Cooperation and Peace (2016), Kwok et al. (2010), and Sol (2021) needs to be effectively addressed to create an environment that enables research and development.

Overall, to create a conducive environment for research in Cambodia and other countries with similar socioeconomic contexts, it is important to build universities' and academics' research capacities (see Chankseliani et al., 2022), without which it is impossible to promote research. It is also crucial to establish clear research policies, both at the institutional and national levels, to provide guidelines on how research can be incentivized, rewarded, or promoted (see Fussy, 2018; Heng et al., 2023b). Without the provision of research incentives or academic recognition in the form of academic promotion or career advancement, it is implausible to expect academics in Cambodia and many other developing countries, distracted and impeded by many challenges and barriers, to devote their time and energy to research. Likewise, if they do not have time to conduct research, given their ultimate attention to teaching and other activities to generate supplementary income (Heng et al., 2023b), it is simply unrealistic to expect research and publication activities coming from such disadvantaged academics. Thus, concerted local and international efforts from different higher education stakeholders are needed to make a difference to the current research and development landscape in Cambodian higher education and elsewhere in the Global South.

5. Conclusion and implications

This study has shown that research and development in Cambodian higher education have experienced positive developments in recent years. Most of the new developments are evident through the introduction of new projects or initiatives that aim to promote the quantity and quality of research and teaching in higher education. There has also been some progress in terms of competitive funding schemes for research made available by MoEYS and some universities. Progress is also seen in relation to an increase in research and publication activities, as several home-grown academic conferences and journals have been established over the past several years. The participants who participated in this study also emphasised the introduction of new courses focusing on research, although such courses are not widely offered across different HEIs.

While progress and new developments can be observed, many challenges to research and development in Cambodian higher education remain. As discussed above, key challenges that this study has found are linked to various inhibiting factors that have been identified in previous studies on higher education and research development in Cambodia (Heng et al., 2022; Sok & Bunry, 2023) and developing world contexts (Fussy, 2019; Horta & Mok, 2020). However, the current study provides new empirical findings that showcase the realities of research and higher education development in Cambodia, particularly at a time when new research-promoting projects and initiatives (e.g., HEIP) have been introduced.

This study bears significance in the sense that it brings to the fore perspectives from a range of higher education stakeholders, including policymakers, educational leaders, lecturers, and students from undergraduate to postgraduate levels. It provides a set of suggestions that are not only useful for promoting research and development in Cambodian higher education but may also be applicable to other similar national contexts. In particular, a number of suggestions, informed by the participants' various perspectives, have been put forward, some of which focus on research capacity building, provision of research incentives and funding, and improvement in the overall national and institutional environment for research. These suggestions add to what has been discussed by previous research, suggesting the need to take into serious consideration the various suggestions informed by the empirical data presented in this study and other recent studies (e.g., Chhaing, 2022; Heng et al., 2022, 2023b). In addition, as this study has shown, there are considerable challenges that constitute a precarious environment for research and development to develop and flourish in Cambodian higher education and other similar contexts. Thus, while concurring with Heng's (2021) proposal for higher education reform in Cambodia, this study calls for new initiatives and collective efforts to accelerate research and development in higher education in the developing world.

This study is not without limitations. Due to the limited sample and the in-depth nature of this qualitative research, extra care should be taken when interpreting the findings presented and discussed in this study. Therefore, it is recommended that future research should include a larger sample size involving other stakeholder groups in higher education in Cambodia. For example, policymakers from the Ministry of Economy and Finance and other relevant ministries or state institutions that play a pivotal role in shaping education and research policies in Cambodia should be approached for key informant interviews. Their perspectives may shed more light on the current research and development landscape in Cambodia.

Moreover, a survey into how university students, teachers, and administrators view the status quo and recent developments in academic research in Cambodia and other similar higher education contexts may be useful in understanding how key higher education stakeholders perceive the current situation and new developments in academic research in their respective countries, and what can be learned from their perspectives to enhance the research and development capacity of HEIs and academics in those countries. Future research may also examine the perspectives of those based in HEIs in provincial or rural areas to promote the voices of these underrepresented stakeholders who are generally not given sufficient attention in previous studies on the development of research and higher education in Cambodia and potentially in other contexts. A deep and nuanced understanding of how relevant actors and stakeholders perceive research and development in Cambodia and how they would like Cambodian higher education to move forward is essential for policy formulation, implementation, and evaluation. This line of research should go beyond the Cambodian context to advance our understanding of how research and development can be promoted in other developing countries.

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Kimkong Heng: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Koemhong Sol:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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